AAEM MINIMONOGRAPH #37

Myokymia is a clinical phenomenon associated with characteristic electromyographic activity referred to as myokymic discharges. These are spontaneously generated bursts of individual motor unit potentials with each burst recurring rhythmically or semirhythmically, usually several times per second. It involves facial muscles more commonly than those of the extremities, and is most often seen in association with Guillain–Barré syndrome, multiple sclerosis, radiation plexopathy, pontine tumors, and timber rattlesnake envenomation. An alteration in the biochemical microenvironment of axon membranes at one of the various sites along the motor axon is the likely basis for the altered membrane excitability that underlies the myokymic discharges in most cases. The similarity of these discharges to those seen with hypocalcemic tetany, and the ability to manipulate myokymic discharges by altering serum-ionized Ca^{++}, suggests that decrease in the ionized Ca^{++} in the microenvironment of the axon may play an important role.
Key words: myokomia • Guillain–Barré syndrome • multiple sclerosis • radiation plexopathy • pontine tumors • rattlesnake venom

MUSCLE & NERVE 14:1043–1049 1991

AAEM MINIMONOGRAPH #37: FACIAL AND LIMB MYOKYMIA

LUDWIG GUTMANN, MD

Myokymia is a clinical phenomenon characterized by undulating, vermicular, rippling, and wavelike movements spreading across the muscle surface. Facial muscles are more commonly affected than those of the extremities. It occurs as a manifestation of a large variety of disorders, most notably Guillain–Barré syndrome,[4,8,21,32] multiple sclerosis,[1,12,13,24,25,33] radiation plexopathy,[14,19,31] brainstem tumors,[9,18,23,25,28,33] and timber rattlesnake (*Crotalus horridus horridus*) envenomation.[5,6] The term "myokymia" appears to have been used first by Schultze in 1894[29] in describing continuous undulating muscle movements of the legs. The phenomenon of facial myokymia was first described by Bernhardt in 1902.[2] Oppenheim associated it with multiple sclerosis in 1916.[24]

Myokymia is associated with characteristic electromyographic activity referred to as myokymic discharges. These are spontaneously generated bursts of individual motor unit potentials with each burst recurring rhythmically or semirhythmically, usually several times per second. Myokymia refers to the *clinical* phenomenon of undulating rippling muscles, and myokymic discharges refer to the associated *electric* activity. Their clinical importance lies in their occurrence with a limited number of disorders and assists in differential diagnostic considerations.

From the Department of Neurology, West Virginia University School of Medicine, Morgantown, West Virginia 26506.

Address reprint requests to American Association of Electrodiagnostic Medicine (formerly the American Association of Electromyography and Electrodiagnosis), 21 Second Street S.W., Suite 306, Rochester, MN 55902.

Accepted for publication November 28, 1990.

CCC 0148–639X/91/01101043–07 $04.00
© 1991 Ludwig Gutmann, MD.
Published by John Wiley & Sons, Inc.

PATHOPHYSIOLOGY

Clinical myokymia is the result of recurring myokymic discharges in multiple neighboring motor units. The exact mechanisms by which these discharges are generated have not been defined but, on the basis of our current knowledge, most cases likely involve a biochemical alteration in the microenvironment of the axon membrane at any one of various sites on the axon between the motor neuron and the terminal axons of the motor unit.[4,5,13] The location of generation sites of the discharges varies with the nature of the underlying disease process.

The disorders that most commonly produce myokymia and myokymic discharges affect the axon membrane microenvironment through such mechanisms as demyelination, radiation changes, direct toxic effects, ischemia, hypoxia, and edema. Axon compression is an uncommon cause of myokymia. Myokymia and myokymic discharges are rarely a feature of compressive nerve lesions

(e.g., carpal tunnel syndrome and ulnar neuropathy).[1,30] Its occasional occurrence with brainstem tumors[9,18,23,25,28,33] and other brainstem masses[3,25] might also argue against the hypothesis of chemical alteration in the motor axon microenvironment. At the same time, the myokymia could reflect the chemical changes due to edema at the intramedullary portion of the axon.

Myokymic discharges arise spontaneously at some portion of the motor axon where there is hyperexcitability of the axon membrane. This may be in the intramedullary portion of the nerve, as in multiple sclerosis,[1,12,13,24,33] or extramedullary (peripherally), as in Guillain–Barré syndrome.[4,8,21,32] The myokymic discharges are typically short bursts of ectopically generated motor unit potentials. The motor unit potentials within bursts most often fire at a rate of 5 to 62 Hz and appear as doublets, triplets, or multiplets (see Figs. 1, 2, 3, and 4).[1,4,21] The recurrently firing potential in a multiplet may fire as often as 128 times per burst (Fig. 5). The terminal spike within the individual bursts may have a prolonged interspike interval.[1] When this interval is sufficiently prolonged, blocking of the terminal spike may occur (Figs. 1 and 2).[4,13] The bursts of motor unit potentials usually fire at 2 to 10 Hz. Occasionally, a burst fires as infrequently as once every 20 seconds. Myokymic bursts with more spikes generally fire at a slower frequency. Bursts fire in a rhythmic or semirhythmic fashion,[21] but the actual rate of recurrence varies from one neighboring motor unit to the next.

The identification of the myokymic discharges may be difficult, especially when recording multiple discharges simultaneously. Multiple myokymic discharges firing at different rates obscure the characteristic sound of a single rhythmic/semirhythmic discharge. In this circumstance, careful manipulation/movement of the recording electrode is necessary to isolate 1 or 2 discharges and to appreciate the rhythmicity of individual recurrent myokymic bursts.

Rasminsky[26] demonstrated similar short recurrent tetanic bursts in nerve roots of dystrophic mice. The excitation of the demyelinated axons was shown to occur as a result of spontaneous ectopic excitation at a demyelinated segment (the ectopic generator); ephaptic excitation involving direct axon to axon nerve action potential transmission (cross talk) at demyelinated segments of 2 axons; or autoexcitation of the ectopic generator by an impulse traversing the demyelinated segment. One or more of these mechanisms may play a role in human myokymic discharges.[26]

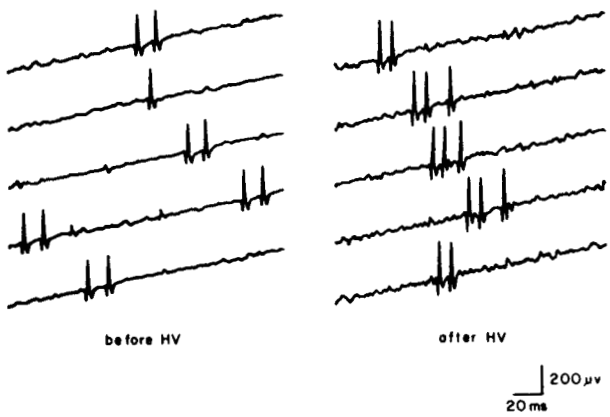


FIGURE 1. A single myokymic discharge (orbicularis oris) seen in successive sweeps (progressing from bottom to top), before and after hyperventilation for 1 minute, from a patient with Guillain–Barré syndrome. Before hyperventilation, the discharge appears as a doublet or singlet. Following hyperventilation, the discharges appear as a triplet or doublet with shorter interspike intervals reflecting burst amplification.

Manipulating Ca^{++} in the microenvironment of the axon alters the firing characteristics of the myokymic discharges. It is tempting to speculate

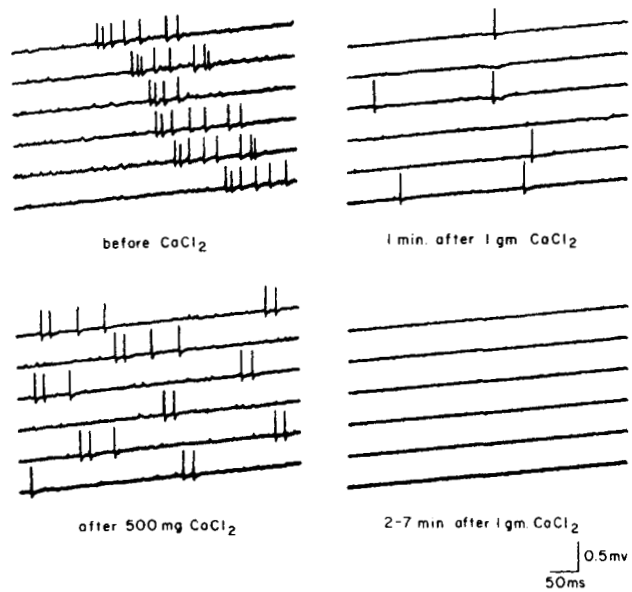


FIGURE 2. A single myokymic discharge appearing as a multiplet (orbicularis oris) seen in successive sweeps (progressing from bottom to top), before and after intravenous $CaCl_2$ (1 g) from another patient with Guillain–Barré syndrome. After the parenteral $CaCl_2$ injection, rapid attenuation of the myokymic bursts occur with decreased number of potentials in each burst and lengthening interspike intervals. Prior to it disappearing, the burst appears as a singlet.

that altered Ca^{++} in the axon microenvironment may play a role in the generation of the discharges as it does in hypocalcemic tetany.[4,5,13] Nerve excitability is altered by changes in ionized Ca^{++} in myelinated and unmyelinated nerves. Lowering the external Ca^{++} concentration lowers the threshold for excitation in these axons; increasing Ca^{++} concentration raises it. Respiratory alkalosis resulting from hyperventilation decreases serum ionized Ca^{++} with a corresponding increase in nerve excitability. Ectopic spike generation in experimentally demyelinated trigeminal nerves increases with hypocapnia, hypocalcemia, or alkalosis. Tetany, due to hypocalcemia, is associated with increased nerve excitability, and spontaneously firing doublets, triplets, and multiplets are recorded from muscle.[10] These discharges are identical to myokymic discharges.[4,5,13]

In Guillain–Barre syndrome, chronic inflammatory polyneuropathy, and timber rattlesnake envenomation, myokymic discharges show burst amplification (increased number of potentials and shortened interspike interval within each burst) due to lowered serum ionized Ca^{++} following hyperventilation or serum-ionized Ca^{++} binding during plasmapheresis (Fig. 1). In contrast, elevating serum-ionized Ca^{++} by intravenous administration of $CaCl_2$, in these disorders, results in prolongation of interspike intervals within the myokymic burst, leading to fewer recurrent action potentials in each burst and, finally, its disappearance (Fig. 2).[4,5] The ectopic generator of the myokymic discharges for each of these disorders is postulated to be in the peripheral portion of the motor axon.

In multiple sclerosis, demyelination occurs in the intramedullary portion of the axon, and this is the most likely site of the myokymic discharge generation. In this circumstance, the response of myokymic discharges to changes in ionized Ca^{++} is absent or diminished. The reason for this is not known but the lack of response may reflect the blood–brain barrier impeding fluxes of ionized Ca^{++}. Other mechanisms could also be factors, including the biochemical difference in peripheral and central myelin and their different cells of origin.

A partial response of the myokymic discharges to increases or decreases in serum-ionized Ca^{++} in some patients with multiple sclerosis occurred in the presence of relatively acute lesions. It is postulated that this is due to a probable acute transient disruption of the blood-brain barrier. In contrast, manipulation of Ca^{++}, when lesions were present

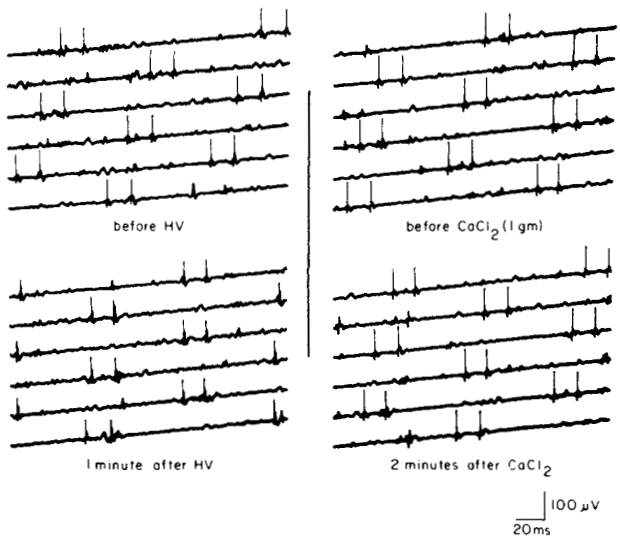


FIGURE 3. A single myokymic discharge appears as a doublet (orbicularis oris) seen in successive sweeps (progressing from bottom to top) showing no response to either 1 minute of hyperventilation and intravenous $CaCl_2$ (1 g). The patient has multiple sclerosis. Myokymia was present for 2 weeks.

from 10 days to 3 years, failed to demonstrate a response at a time when an alteration in the blood-brain barrier would be expected to be less severe or absent (Fig. 3).[13]

Local nerve blocks result in a gradual cessation of myokymic discharges over several minutes if placed distal to the location of the generator. In the case of 2 patients with radiation-induced brachial and lumbosacral plexopathies, 2% lidocaine blocks placed in the ulnar nerve at the elbow and peroneal nerve at the knee, respectively, resulted in myokymic discharges disappearing within 6 minutes.[1]

In rattlesnake envenomation, the facial myokymic results from hematogenous spread of venom from the bitten extremity (Fig. 4). The rapid disappearance of facial myokymia, that occurs within several hours after the administration of antivenin, lends further support to the hypothesis that biochemical changes within the microenvironment of the axon result in the altered nerve excitability.[5]

CLINICAL FEATURES

Myokymia is most frequently seen in Guillain–Barré syndrome,[4,8,21,32] multiple sclerosis,[1,12,13,24,33] and radiation plexitis.[14,19,31] It occurs as an invariable feature in the face and bitten extremity in timber rattlesnake envenomation.[5,6] Facial myokymia occasionally occurs with in-

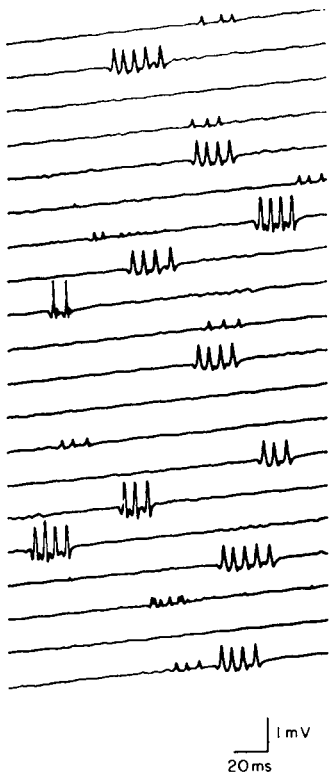


FIGURE 4. A myokymic discharge appears as a triplet/multiplet (gastrocnemius muscle) recurring in successive sweeps (progressing from bottom to top). The patient had been bitten in the ipsilateral ankle by a timber rattlesnake 2 days earlier. Two other myokymic discharges are seen firing at different rates in the background.

tramedullary pontine tumors (primary or metastatic)[9,18,23,25,28,33] and tuberculoma,[3] cerebellar and cerebellopontine angle tumors,[9,25] meningeal carcinomatosis/sarcomatosis,[25] syringobulbia,[25] lymphocytic meningoradiculitis,[11] basilar invagination,[7] phosgene poisoning,[25] and following cardiopulmonary arrest.[22] In addition to radiation plexitis and timber rattlesnake envenomation, limb myokymia occurs occasionally in multiple sclerosis,[1] acute and chronic inflammatory polyneuropathies,[1,4,21] ischemic neuropathy,[1] and occasional compression neuropathies.[1,30]

Facial myokymia and/or myokymic discharges occur in 15% of patients with Guillain–Barré syndrome.[21] The degree of clinical myokymia ranges from obvious to imperceptible. Myokymia in the extremities occurs but is much less frequent.[1,21] In the Guillain–Barré syndrome, the facial myokymia/myokymic discharges are frequently bilateral and often associated with mild facial weakness. It appears within the first 3 weeks of the illness, and is present up to 3 weeks before clearing. The limb myokymia in this disorder is also transient but may persist for as long as 6 weeks.[21]

Table 1. Myokymic disorders.

- Multiple sclerosis
- Guillain–Barré syndrome
- Chronic inflammatory polyneuropathy
- Radiation plexopathy
- Intramedullary/extramedullary pontine tumors/masses
- Neoplastic/inflammatory meningoradiculitis
- Anoxic/ischemic rhombencephalopathy
- Timber rattlesnake envenomation

In multiple sclerosis, myokymia and/or myokymic discharges also involve facial muscles much more frequently than those of the extremities. In the face, it occurs unilaterally and is transient, usually present for only a few weeks, but it may persist for up to 3 months. Recurrent episodes occasionally occur and may be either on the same or opposite sides. It is not associated with any significant facial weakness. Its presence implies a pontine lesion, but pathologic confirmation of the exact lesion location is lacking.[12,13]

The occasional segmental occurrence of myokymia in the extremities of patients with multiple sclerosis is associated with other clinical evidence of myelopathy. In 1 autopsied case, a demyelinating plaque was present in the ventral portion of the lower cervical cord when myokymia had been previously observed in the first dorsal interosseous muscle.[1] This finding lends further support to the hypothesis that the myokymic discharges arise from an ectopic generator within the motor unit axon.

Facial myokymia in the presence of a posterior fossa tumor or mass is uncommon but more likely to occur with the mass located within the brainstem rather than outside.[9] The myokymia has oc-

Table 2. Myokymic discharges characteristic features.

- Short burst of a motor unit potential (5 to 62 Hz)
- Bursts appear as doublets, triplets, multiplets
- Bursts recur rhythmically/semirhythmically (usually 2 to 10 Hz)
- Facial more common than extremity muscles
- Transient (days–weeks) in inflammatory polyneuropathies, multiple sclerosis and timber rattlesnake envenomation
- Persistent in radiation plexopathy and pontine tumors
- Usually unilateral in multiple sclerosis, pontine tumors, radiation plexopathy
- Bilateral in inflammatory polyneuropathies, anoxia/ischemia, timber rattlesnake envenomation

curred in association with pontine gliomas, cerebellar astrocytomas, metastatic tumors, and acoustic neuromas. The myokymia, in this circumstance, is unilateral and usually persistent.[9,18,23,25,28,33] Autopsies have shown sparing of the ipsilateral facial nucleus by tumor while edema involved the area of the nucleus.[23,28,33] This may suggest that factors other than compression are responsible for the myokymia. Myokymia may rarely occur in muscles innervated by other cranial nerves in the presence of a brainstem tumor.[18]

Myokymia is a common occurrence with brachial and lumbosacral plexopathy due to radiation (Fig. 5).[1,14,19,31] In 3 series, it occurred in 60% to 70% of cases. The myokymia is present in only one or a few muscles of the involved extremity, and these muscles are usually only mildly or moderately weak.[1,14,19,31] The myokymia may be present decades after the radiation occurred. It is often associated with fibrillations and neurogenic motor unit potential changes, but the presence of the myokymia helps to distinguish radiation from tumor infiltration or traumatic plexopathy. Myokymia only occasionally occurs in the latter 2 circumstances. In another series of myokymias of various causes limited to the limbs, 70% obtained it from radiation-induced plexopathy and/or radiculopathy, or radiation myelopathy. Facial myokymia due to radiation has also been reported.[1] Thus, myokymia is a reliable marker for radiation change.[1]

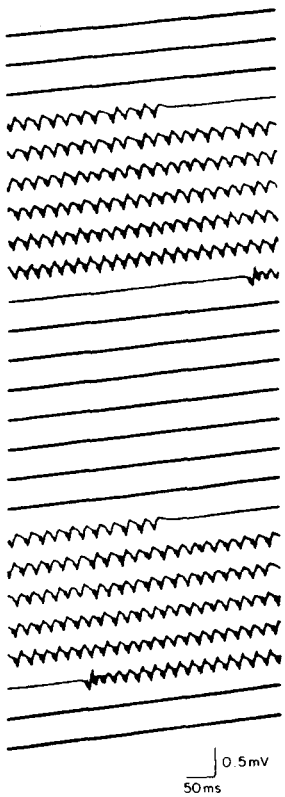


FIGURE 5. A single myokymic discharge appearing as a prolonged multiplet (flexor carpi radialis) seen twice in successive sweeps (progressing from bottom to top) in a patient with radiation-induced brachial plexopathy.

Timber rattlesnake envenomation is invariably associated with bilateral facial myokymia and myokymia of the bitten extremity.[5] The facial myokymia is a remote effect, due to the hematogenous spread of the toxin. It disappears within several hours of the patient receiving antivenin therapy. The myokymia in the bitten hand or leg persists 3 days before clearing and is, in part, manifested by a characteristic abduction/adduction tremor of the fingers and a vertical tremor of the toes.[6]

Facial myokymia has been reported in association with Bell's palsy and hemifacial spasm,[25] but this must be extraordinarily unusual. Its occurrence with facial palsy is much more likely to reflect underlying generalized inflammatory neuropathy, multiple sclerosis, or mass in or adjacent to the pons.

A syndrome of continuous motor unit activity, impaired muscle relaxation, and hyperhidrosis has been described.[4,16,20] This is a rare disorder sometimes referred to as Isaac's syndrome.[17] Much of the spontaneous motor unit activity often occurs in prolonged trains at rates of 150 to 300 Hz; however, in some cases, myokymic discharges are the predominant abnormality.[4,16] The syndrome has been noted as part of chronic inflammatory polyneuropathy.[4]

THERAPY

In most circumstances, myokymia is asymptomatic and requires no therapy. At times it may not be apparent to the patient until noted by the physician. Occasionally, generalized limb myokymia may be of such prominence in an inflammatory polyneuropathy to warrant therapy.[4] Isaac's syndrome results in similar muscle stiffness.[16,20] In these circumstances, phenytoin and carbemazepine in anticonvulsant dosages have been successful in reducing the amount of myokymic discharges.[4,16,27]

DIFFERENTIAL DIAGNOSIS

The reader may find the AAEE Glossary of Terms in Clinical Electromyography (*Muscle & Nerve* 1987; 10[suppl 8]) a good supplement for

this section. Recurrently firing doublets and triplets are a characteristic physiologic finding, with hypocalcemic tetany reflecting the decreased serum Ca^{++}. It is always important to consider tetany in the differential diagnosis of myokymia.[10,27]

Myokymia differs from fasciculations both clinically and electrophysiologically. Fasciculations are brief twitches of a group of muscle fibers that represent spontaneous single discharges of individual motor units. They appear as single random motor unit potentials. An occasional fasciculation may fire as a doublet, but not recurrently.

Synkinesis, seen with hemifacial spasm and as a late sequelae of Bell's palsy, appears distinctly different clinically with synchronous contraction of many motor units in facial muscles. However, electromyographically, it can be confused with myokymia, especially when only 1 or a few motor unit potentials are being recorded. In this circumstance, the bursts can appear rhythmic or semirhythmic, as when recording from the orbicularis oris and the patient blinking his eyes rhythmically following a Bell's palsy.

Blepharospasm is a prolonged tonic contraction of the orbicularis oculi muscles and may be clinically confused with myokymia. However, the electromyographic features of blepharospasm, are similar to those of multiple motor unit potential activity during a muscle contraction.

Neuromyotonic discharges are prolonged bursts of motor unit potentials firing at rates of 150 to 300 Hz for a few seconds, often beginning and ending abruptly. The potential amplitude may wane. Unlike myokymia, the bursts do not occur repetitively in a rhythmic fashion and may be initiated by needle movement, voluntary effort, or nerve percussion.

A complex repetitive discharge (CRD) is a burst of a recurrently firing complex muscle action potential that begins spontaneously or after needle electrode movement. CRDs begin and end abruptly and often last several seconds, firing at a frequency of 5 to 100 Hz. CRDs do not recur in a rhythmic fashion.

Myotonic discharges occur as a burst of a biphasic spike potential or positive wave at rates of 20 to 80 Hz. They occur in response to needle electrode movement, voluntary contraction, or muscle percussion. They typically have a waxing and waning quality, with an associated change in pitch, giving it the sound of a dive bomber. Bursts do not recur spontaneously in a rhythmic fashion.

The term "superior oblique myokymia" is a misnomer, since myokymic discharges do not produce the intermittent regular contractions of the superior oblique muscle. The periodic intorsion of the affected eye is due to intermittent rapid phasic firing of involved motor units without reciprocal phasic inhibition in the antagonist muscle.[15] The latter disorder is more analagous to hemifacial spasm.

REFERENCES

1. Albers JW, Allen AA, Bastron JA, Daube JR: Limb myokymia. *Muscle Nerve* 1981;4:494–504.
2. Bernhardt M: Ein ungewoehnlicher Fall von Facialkrampf. *Neurol Zbl* 1902;21:689–695.
3. Boghen D, Filiatrault R, Descarries L: Myokymia and facial contracture in brain stem tuberculoma. *Neurology* 1977;27:270–272.
4. Brick JF, Gutmann L, McComas CF: Calcium effect on generation and amplification of myokymic discharges. *Neurology* 1982;32:618–622.
5. Brick JF, Gutmann L, Brick J, Apelgren KN, Riggs JE: Timber rattlesnake venom-induced myokymia: Evidence for peripheral nerve origin. *Neurology* 1987;37:1545–1446.
6. Brick JF, Gutmann L: Abduction–adduction hand tremor of myokymia. *Neurology* 1988;38:995.
7. Chalk CH, Litchy WJ, Ebersold MJ, Kispert DB: Facial myokymia and unilateral basilar invagination. *Neurology* 1988;38:1811–1812.
8. Daube JR, Kelly JJ, Martin RA: Facial myokymia with polyradiculo-neuropathy. *Neurology* 1979;29:662–669.
9. Espinoza RE, Lambert EH, Klass DW: Facial myokymia affecting the electroencephalogram. *Mayo Clin Proc* 1967;42:258–270.
10. Feldman RG, Granger CV: Electromyography in latent tetany. *N Engl J Med* 1963;269:1064–1067.
11. Gemignani F, Juvarra G, Calzetti S: Facial myokymia in the course of lymphocytic meningoradiculitis: Case report. *Neurology* 1981;31:117–118.
12. Gutmann L, Thompson HG, Martin JD: Transient facial myokymia. *JAMA* 1969;209:389–390.
13. Gutmann L, Brick JF, Riggs JE: Calcium myokymia of brainstem origin. *Neurology* 1986;36:126–130.
14. Harper CM, Thomas JE, Cascino TL, Litchy WJ: Distinction between neoplastic and radiation-induced brachial plexopathy, with emphasis on the role of EMG. *Neurology* 1989;39:502–506.
15. Hoyt WF, Keane JR: Superior oblique myokymia. *Arch Ophthalmol* 1970;84:561–467.
16. Hughes RC, Matthews WB: Pseudomyotonia and myokymia. *J Neurol Neurosurg Psychiatry* 1969;32:11–14.
17. Isaacs H: A syndrome of continuous muscle fibre activity. *J Neurol Neurosurg Psychiatry* 1961;24:319–325.
18. Kaeser HE, Skorpil V: Myokymia involving muscles innervated by the Vth, VIth, VIIth, IXth, Xth, XIIth cranial nerves with brain stem tumor. *Eur Neurol* 1976;14:408–412.
19. Levin KH, Lederman RJ, Wilborn AJ: Spectrum of EMG changes in radiation brachial plexopathy. *Muscle Nerve* 1987;10:656.
20. Lublin FD, Tsairis P, Streletz LJ, Chambers RA, Riker WF, Van Poznak A, Duckett SW: Myokymia an impaired mus-

cular relaxation with continuous motor unit activity. *J Neurol Neurosurg Psychiatry* 1979;42:557–562.
21. Mateer JE, Gutmann L, McComas CF: Myokymia in Guillain–Barré syndrome. *Neurology* 1983;33:374–376.
22. Morris HH, Estes ML: Bilateral facial myokymia following cardiopulmonary arrest. *Arch Neurol* 1981;38:393–394.
23. Negri S, Caraceni T, DeLorenzi L: Facial myokymia and brain stem tumor. *Eur Neurol* 1976;14:108–118.
24. Oppenheim H: Neue Beitrage zur Klinik der multiplen Sklerose. *Jahresbericht Neurologie Psychiatrie* 1916;20:14–18.
25. Radu EW, Skorpil V, Kaeser HE: Facial myokymia. *Eur Neurol* 1975;13:499–512.
26. Rasminsky M: Ephaptic transmission between single nerve fibers in the spinal nerve roots of dystrophic mice. *J Physiol (Lond)* 1980;305:151–169.
27. Schaaf M, Payne C: Effect of diphenylhydantoin and phenobarbital on overt and latent tetany. *N Engl J Med* 1966;274:1228–1233.
28. Sethi PK, Smith BH, Kalyanaraman K: Facial myokymia: A clinicopathological study. *J Neurol Neurosurg Psychiatry* 1974;37:745–749.
29. Shultze F: Beitrage zur muskelpathologie. *Deutsch Z Nervenheilk* 1894;6:65–75.
30. Streib EW: Distal ulnar neuropathy as a cause of finger tremor: A case report. *Neurology* 1990;40:153–154.
31. Thomas JE, Cascino TL, Earle JD: Differential diagnosis between radiation and tumor plexopathy of the pelvis. *Neurology* 1985;35:1–7.
32. Wasserstrom WR, Starr A: Facial myokymia in the Guillain–Barré syndrome. *Arch Neurol* 1977;34:576–577.
33. Waybright EA, Gutmann L, Chou SM: Facial myokymia pathological features. *Arch Neurol* 1979;36:244–245.